Adam: A Method for Stochastic Optimization

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Abstract

We introduce Adam, an algorithm for first-order gradient-based optimization of stochastic objective functions, based on adaptive estimates of lower-order moments. The method is straightforward to implement, is computationally efficient, has little memory requirements, is invariant to diagonal rescaling of the gradients, and is well suited for problems that are large in terms of data and/or parameters. The method is also appropriate for non-stationary objectives and problems with very noisy and/or sparse gradients.

1. Introduction

Stochastic gradient-based optimization is of core practical importance in many fields of science and engineering. Many problems can be cast as the optimization of an expected value of an objective function with respect to parameters theta. Adam combines the advantages of two popular methods: AdaGrad (which works well with sparse gradients) and RMSProp (which works well in on-line and non-stationary settings).

Algorithm & Moment Equations

Section 2: Mathematical Formulation of First and Second Moments

2.1 Moving Averages and Bias Correction

Let g_t be the gradient at step t. The moving averages of the gradient m_t and squared gradient v_t are updated with decay rates beta_1 and beta_2:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

2.2 Bias-Corrected Estimates and Parameter Update

Since m_t and v_t are initialized as zero vectors, they are biased toward zero, especially during initial steps. Bias-corrected estimates hat{m}_t and hat{v}_t are computed as hat{m}_t = m_t / (1 - beta_1^t) and hat{v}_t = v_t / (1 - beta_2^t). The parameter update rule is:

$$\theta_t = \theta_{t-1} - \frac{\alpha \cdot \hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Default Hyperparameters & Empirical Convergence

Section 4: Recommended Default Settings and Empirical Results

4.1 Default Hyperparameters Table

Recommended standard hyperparameter configurations across machine learning problems:

Hyperparameter	Symbol	Default Value	Description
Learning Rate	alpha	0.001	Step size / scaling factor
First Moment	beta_1	0.9	Exponential decay for first moment
Second Moment	beta_2	0.999	Exponential decay for second moment
Numerical Stab	epsilon	1e-8	Division by zero prevention

5. Limitations

Adam requires maintaining two additional state vectors (m_t and v_t) for each parameter in the model, tripling the memory requirement for model weights during optimization. Furthermore, in certain non-convex optimization problems, the learning rate schedule can fail to converge, motivating subsequent modifications such as AMSGrad and AdamW with decoupled weight decay.